Solution exercise set 4

Problem 1:

a) See the R-code file.

b) See the R-code file.

c) We use the property that the number of arrivals in a Poisson process over a time interval of length t is Poisson distributed with expectation $\lambda t = 2 \cdot 10 = 20$. Let $X = N(10)$, then

$$P(X > 25) = 1 - P(X \leq 25) = 1 - \sum_{x=0}^{25} \frac{20^x}{x!} \exp(-20) = 1 - \texttt{ppois}(25, 20) = \underline{\underline{0.112}}$$

where `ppois` is the R function to compute cumulative Poisson probabilities.

Rizzo, exercise 3.1:

Two parameter exponential cumulative distribution function:

$F(x) = P(X \leq x) = \int_{\eta}^{x} \lambda e^{-\lambda(y-\eta)} dy = 1 - e^{-\lambda(x-\eta)}, x \geq \eta.$
This is simply an exponential distribution shifted η to the right.

Simulation, two possibilities:

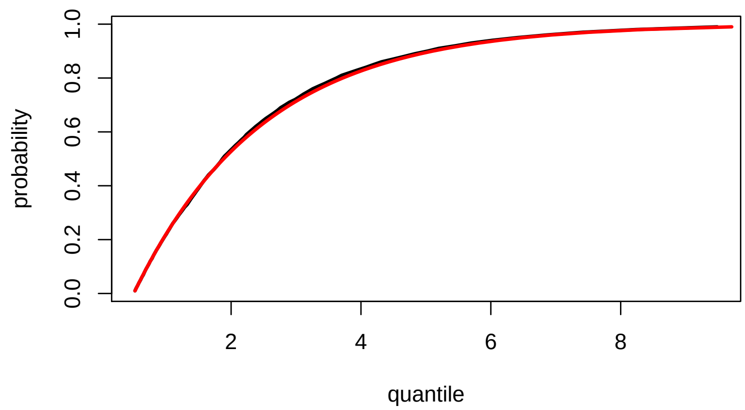
1. If $Y \sim exp(\lambda)$, then $Y + \eta \sim exptwo(\lambda, \eta)$.
   I.e.: simulate and $exp(\lambda)$-variable and add η.

2. By using the inverse transform algorithm:
   $U = F(X) \Rightarrow X = F^{-1}(U) = \dfrac{\ln(1-U)}{-\lambda} + \eta \quad \sim \dfrac{\ln(U)}{-\lambda} + \eta$
   I.e.: simulate an $U(0,1)$-variable and calculate $\frac{\ln(1-U)}{-\lambda} + \eta$ (or $\frac{\ln(U)}{-\lambda} + \eta$, since U and $1-U$ have the same distribution).

Quantile: The q-quantile is defined as the x-value having the property that $P(X \leq x) = q$. Thus the theoretical quantile is determined by the equation: $q = F(x)$, and this can be solved for x: $q = 1 - e^{-\lambda(x-\eta)} \Rightarrow x = \dfrac{\ln(1-q)}{-\lambda} + \eta$.

See the R-code file for the implementation.

Probabilities vs. quantiles - black is estimated, red is theoretical:



Rizzo, exercise 3.11:

See the R-code file. It seems like values of p_1 in the interval [0.25, 0.75] produces a bimodal distribution.

Rizzo, exercise 4.1:

See the R-code file. Notice that $S_n = 10+Y_1+Y_2+\cdots+Y_n$ where $Y_1, Y_2, \ldots$ are independent with $P(Y=-1) = P(Y=1) = 0.5$. The process terminates when S_n for the first time hit 0 or 20.